

ENGR 421/DASC 521: Introduction to Machine Learning

Homework 06: Area Under the ROC Curve

Deadline: January 14, 2026, 11:59 PM

In this homework, you will implement the receiver operating characteristics (ROC) curve in Python. Here are the steps you need to follow:

1. Read Section 20.7 from the textbook.
2. You are given the true labels of 1000 data points for a binary classification problem in the file named `hw06_true_labels.csv`, the predicted posterior probabilities of these 1000 data points for the positive class obtained using **Algorithm 1** in the file named `hw06_predicted_probabilities1.csv`, and the predicted posterior probabilities of these 1000 data points for the positive class obtained using **Algorithm 2** in the file named `hw06_predicted_probabilities2.csv`.
3. Calculate possible classification thresholds using the predicted posterior probabilities. (20 points)

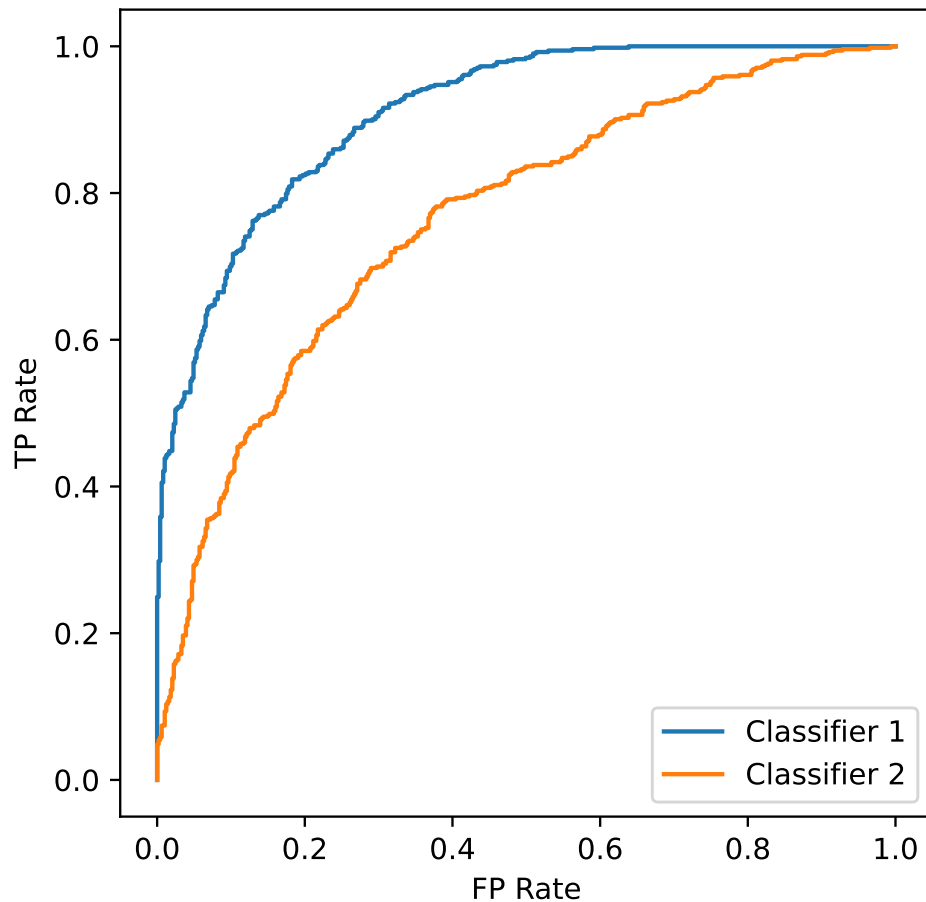
```
print(thresholds1)
[0.0118935  0.0264396  0.03009405 ... 0.982222  0.9899465  0.996494  ]
```

```
print(thresholds2)
[0.00108003 0.00231496 0.00427099 ... 0.998449  0.9988205  0.999556  ]
```

4. Calculate FP and TP rates using the true labels, predicted posterior probabilities, and thresholds. (40 points)

```
print(fp_rates1[495:505])
print(tp_rates1[495:505])
[0.18275154 0.18275154 0.18069815 0.17864476 0.17864476 0.17864476
 0.17659138 0.17659138 0.17659138 0.17659138]
[0.81091618 0.80896686 0.80896686 0.80896686 0.80701754 0.80506823
 0.80506823 0.80311891 0.80116959 0.79922027]
```

```
print(fp_rates2[495:505])
print(tp_rates2[495:505])
[0.29979466 0.29774127 0.29774127 0.29568789 0.2936345  0.29158111
 0.28952772 0.28952772 0.28952772 0.28747433]
[0.69980507 0.69980507 0.69785575 0.69785575 0.69785575 0.69785575
 0.69785575 0.69590643 0.69395712 0.69395712]
```



5. Calculate the area under the ROC curve using the FP and TP rates. (40 points)

```
print("The area under the ROC curve for Algorithm 1 is {}".format(auroc1))
The area under the ROC curve for Algorithm 1 is 0.9091185641493649.
print("The area under the ROC curve for Algorithm 2 is {}".format(auroc2))
The area under the ROC curve for Algorithm 2 is 0.7640404913721685.
```

What to submit: You need to submit your source code in a single file (.py file). You are provided with a template file named as 0099999.py, where 99999 should be replaced with your 5-digit student number. You are allowed to change the template file between the following lines.

```
# your implementation starts below

# your implementation ends above
```

How to submit: Submit the file you edited to LearnHub by following the exact style mentioned. Submissions that do not follow these guidelines will not be graded.

Late submission policy: Late submissions will not be graded.

Cheating policy: Very similar submissions will not be graded.
